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## TEST011 - Performance Testing Report – Food Remedy Pipeline

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### 1. Objective

The objective of this test was to evaluate the performance and stability of the Food Remedy data pipeline when processing a large dataset (50,000+ records). The focus was on measuring execution time, throughput, and system reliability under load.

### 2. Test Environment

- Platform: Windows (Local Machine)
- Runtime: Python 3.14.5
- Execution Mode: Local CLI Pipeline (No Firebase / Dry-run enabled for testing stability)
- Pipeline Mode: Enrichment Stage Active (Clean & Seed disabled)

### 3. Dataset Details

- Total dataset size: **61,373 records**
- Data split into chunks: **7 chunks**
- Chunk size: **~10,000 records per batch**

### 4. Execution Summary

| Metric                  | Value                 |
|-------------------------|-----------------------|
| Total Records Processed | 61,373                |
| Successful Records      | 61,373                |
| Failed Records          | 0                     |
| Failed Chunks           | 0                     |
| Total Execution Time    | 116.31 seconds        |
| Average Chunk Time      | ~14–20 seconds        |
| Throughput              | 527.69 records/second |
| Peak Memory Usage       | 174.08 MB             |
| System Stability        | Stable                |

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### 5. Performance Results

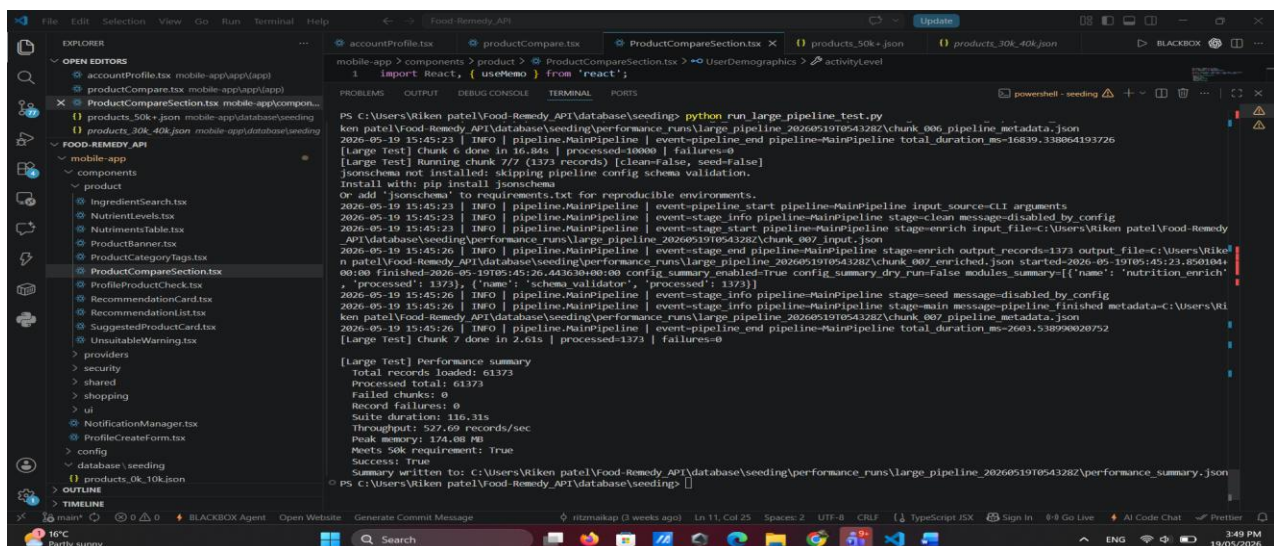
- The pipeline processed all 61,373 records successfully without any crashes or interruptions.
- Each batch of 10,000 records was processed consistently within ~14–20 seconds.
- The final partial batch (1,373 records) completed in 2.61 seconds, showing efficient scaling for smaller loads.
- The system maintained stable memory usage (peak ~174 MB), indicating efficient resource handling.

### 6. Key Observations

- ✓ No data loss or processing failures occurred.
- ✓ Performance remained consistent across all chunks.
- ✓ Pipeline demonstrated strong scalability for large datasets.
- ✓ No system crashes or memory overload detected.
- ✓ Enrichment stage was the primary active processing step.

### 7. Conclusion

The Food Remedy pipeline successfully handled a large dataset of 61,373 records with stable performance and no failures. The system achieved a throughput of approximately 527 records per second, confirming that it meets the required performance criteria for large-scale data processing.



```
PS C:\Users\Riken patel\Food-Remedy\API\database\seeding> python run_large_pipeline_test.py
ken patel\Food-Remedy\API\database\seeding\performance_runs\large_pipeline_2026051910543282\chunk_006.pipeline.metadata.json
2026-05-19 15:45:23 | INFO | pipeline.MainPipeline | event-pipeline_end pipeline-mainPipeline total_duration_ms=16839.338064191726
[Large Test] Chunk 6 done in 16.84s | processed=10000 | failures=0
[Large Test] Running chunk 7 (1373 records) [clean=false, seed=false]
Jsonschema not installed: skipping pipeline config schema validation.
Install with: pip install jsonschema
Or add 'jsonschema' to requirements.txt for reproducible environments.
2026-05-19 15:45:23 | INFO | pipeline.MainPipeline | event-pipeline_start pipeline-mainPipeline input_source=CLI arguments
2026-05-19 15:45:23 | INFO | pipeline.MainPipeline | event-stage_info pipeline-mainPipeline stage-clean message-disabled by config
2026-05-19 15:45:23 | INFO | pipeline.MainPipeline | event-stage_start pipeline-mainPipeline stage-enrich input_file=C:\Users\Riken patel\Food-Remedy
API\database\seeding\performance_runs\large_pipeline_2026051910543282\chunk_007_input.json
2026-05-19 15:45:26 | INFO | pipeline.MainPipeline | event-stage_end pipeline-mainPipeline stage-enrich output_records=1373 output_file=C:\Users\Riken
n patel\Food-Remedy\API\database\seeding\performance_runs\large_pipeline_2026051910543282\chunk_007_enriched.json started=2026-05-19T05:45:23.850104+
00:00 finished=2026-05-19T05:45:26.443309+00:00 config_summary_enabled=true config_summary_dry_run=false modules_summary=[{'name': 'nutrition_enrich'
, 'processed': 1373}, {'name': 'schema_validator', 'processed': 1373}]
2026-05-19 15:45:26 | INFO | pipeline.MainPipeline | event-stage_info pipeline-mainPipeline stage-seed message-disabled by config
2026-05-19 15:45:26 | INFO | pipeline.MainPipeline | event-stage_info pipeline-mainPipeline stage-main message-pipeline_finished metadata=C:\Users\Ri
ken patel\Food-Remedy\API\database\seeding\performance_runs\large_pipeline_2026051910543282\chunk_007.pipeline.metadata.json
2026-05-19 15:45:26 | INFO | pipeline.MainPipeline | event-pipeline_end pipeline-mainPipeline total_duration_ms=2603.538990020752
[Large Test] Chunk 7 done in 2.61s | processed=1373 | failures=0

[Large Test] Performance summary
Total records loaded: 61373
Processed total: 61373
Failed chunks: 0
Record failures: 0
Suite duration: 116.31s
Throughput: 527.60 records/sec
Peak memory: 174.08 MB
Meets 50k requirement: True
Successful: True
Summary written to: C:\Users\Riken patel\Food-Remedy\API\database\seeding\performance_runs\large_pipeline_2026051910543282\performance_summary.json
PS C:\Users\Riken patel\Food-Remedy\API\database\seeding>
```